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Challenging the Horizontal Assumption: Tilted Fluid Contacts in a Complex Carbonate Field

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Abstract

Tilted fluid contacts refer to oil-water or gas-water interfaces that are not level or horizontal across the reservoir. These deviations can result from structural gradients, capillary pressure variation, or hydrodynamic flow and introduce substantial complexity into reservoir characterization. In carbonate settings, this complexity is further amplified due to inherent heterogeneities in rock properties and subtle structural features. Unlike conventional models that assume fluid level equilibrium, tilted contacts represent spatial variations that must be interpreted with a geologically consistent framework. Identifying and understanding such contacts is critical for accurate reserves estimation, optimal well placement, and robust static and dynamic modeling (Kirkham et al., 2010).

This case study focuses on Field G, a mature carbonate reservoir in the Middle East, where spatial variation in oil-water contact (OWC) depths across the Central and Eastern areas triggered an investigation into the potential presence of a tilted contact. The study utilizes a comprehensive dataset, integrating petrophysical analysis, core-based capillary pressure measurements, and 3D seismic interpretation. Although pressure data were reviewed, they did not provide conclusive evidence of a consistent Free Water Level (FWL), reinforcing reliance on log-based interpretations. The study evaluates the interplay of facies, permeability, and structural features and their collective role in governing contact geometry. The resulting observations suggest that a facies-controlled tilted contact, possibly influenced by long-term structural tilting, is a plausible explanation for the observed OWC variation.

This integrated analysis enhances understanding of the reservoir's architecture and highlights the need to revisit assumptions regarding fluid level equilibrium in similar carbonate systems. The implications are significant for reserves estimation, well planning, and full-field development strategies in mature, heterogeneous carbonate settings.

Introduction

In conventional reservoir models, fluid contacts are assumed to be horizontal, governed by equilibrium between capillary pressure and fluid density. However, in complex carbonate systems, variations in rock properties, structural geometry, and diagenetic history may lead to deviations from this idealized scenario.

In Field G, a prolific carbonate reservoir located in the Middle East, fluid contact variations have been observed through successive drilling campaigns, particularly between the Central and Eastern sectors. The observed discrepancy in OWC depths of up to 40 meters, combined with an absence of structural compartmentalization, raised questions regarding the nature of the contact and the appropriateness of applying a flat contact model.

This paper investigates whether a tilted contact, governed by facies variation and subtle structural trends, better explains the observed OWC pattern. The approach integrates petrophysical evaluation, sedimentological insights, core-based saturation data, and seismic interpretation to explore the possibility of a non-horizontal, facies-controlled contact surface. The implications of this interpretation are significant, influencing reservoir modeling, volumetrics, well placement, and long-term development planning.

Field Overview

Situated in the Middle East, Field G is a large carbonate oil reservoir with over 146 wells drilled to date. The field was initially discovered in the late 1980s through exploration well X-1. This was followed by wells X-2 and X-3, which were drilled by the Host Authority of Country G to delineate the reservoir's extent. A petroleum development agreement was later established among Company A, Company B, and the Host Authority, appointing Company A as the field's operator. Since then, Company A has carried out drilling operations for more than 146 wells with no major incidents.

Production is currently focused on Reservoir X, the field's main carbonate reservoir, which exhibits an average porosity of around 20%. All production wells have been completed as single-selective producers. Several secondary reservoir targets—comprising both carbonate and clastic formations—remain undeveloped. Although the field is not known to contain major faults, repeated discrepancies between expected and actual OWC depths during drilling led to a full-field reevaluation. A systematic deviation in contact depths suggested the need to reassess previous assumptions regarding fluid contact geometry and consider more complex geological and petrophysical controls.

Petrophysical Analysis

A comprehensive petrophysical evaluation was undertaken to characterize fluid contact behavior and support the hypothesis of a tilted OWC across Field G. This analysis integrated conventional log interpretation, core calibration, and saturation modeling to assess vertical and lateral variations in fluid distribution. Key log types, including deep and shallow resistivity, neutron, density, and sonic logs, were systematically reviewed to identify fluid contacts, interpret facies changes, and derive petrophysical properties such as porosity, water saturation, and permeability proxies.

The resistivity logs played a central role in identifying the oil-water contact (OWC), particularly where the transition was sharp. In most wells across the Central Area, the OWC was encountered at shallower depths, typically 35-40 meters higher than in the Eastern Flank (Figure 1). Resistivity log overlays (Figure 2) were utilized to define the contact depths. Laterolog-type resistivity tools were employed, providing both Micro Resistivity (R_{micro}) and Deep Resistivity (R_{deep}) curves. Due to the high salinity contrast between the formation water and the drilling mud—exceeding 100,000 ppm—the Oil-Water Contact (OWC) can be reliably identified at the crossover point between R_{micro} and R_{deep} (Figure 2). The reliability of these interpretations was enhanced through calibration with formation tester data, from which two water samples were obtained from centrally located wells below the interpreted OWC. Furthermore, oil and water samples collected from the Eastern Area confirmed the presence of a deeper OWC in that region.

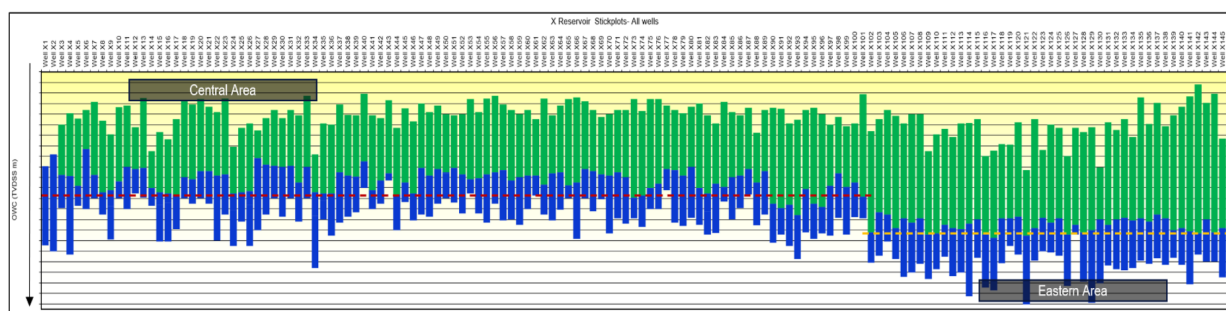


Figure 1—Stick Plot Showing Shallower OWC at Central Area Compared to Eastern Area

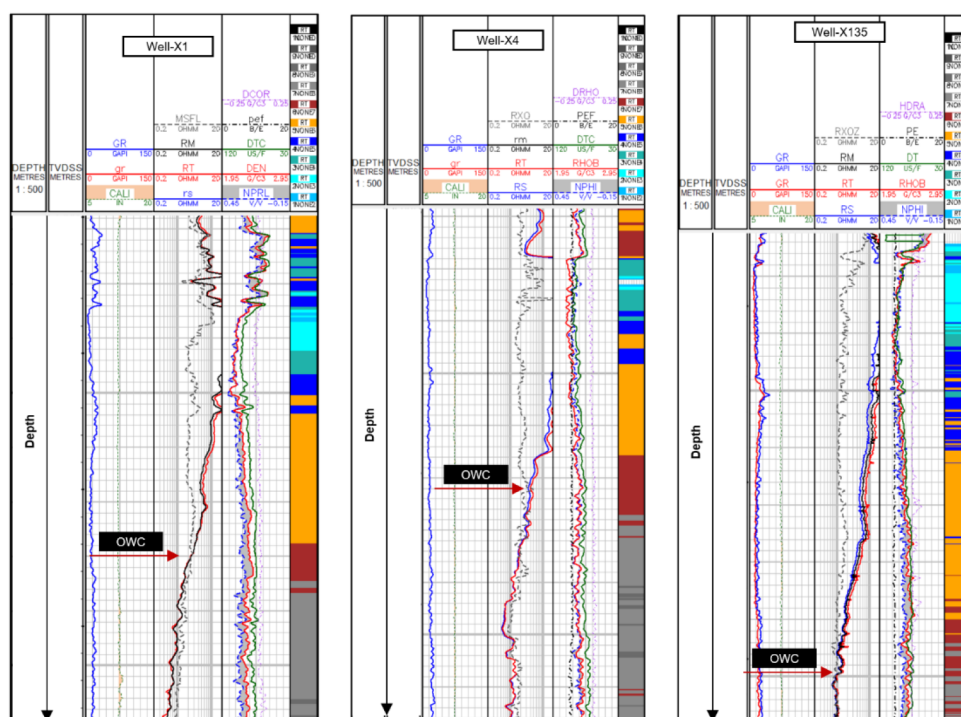


Figure 2—Resistivity Overlay to Determine OWC

To further validate the OWC distribution, a series of saturation-height functions (SHFs) were derived from laboratory-measured capillary pressure curves. These SHFs, based on core plugs representing various depositional facies, were used to generate log-based water saturation profiles (Kumar et al., 2002). Differences in the height above FWL (HAFL) saturation profiles across the field provided additional evidence for lateral variability in the capillary behavior of the reservoir (Figure 3). In particular, the Central Area consistently displayed higher capillary entry pressures and tighter pore throat distributions, a finding that correlated well with the facies identified through core description and thin-section petrography.

Pressure data were initially expected to offer a means to independently define a free water level (FWL) by constructing pressure-depth plots. However, the application of this method was limited. Many wells lacked sufficient pressure measurements in the water leg, and the few that did show scatter or inconsistency, likely due to heterogeneities in permeability and borehole conditions. Moreover, the low mobility of the formation water, due to high water saturations and low permeability zones, likely contributed to the poor pressure gradient definition. Consequently, pressure-based FWL analysis was inconclusive, and the interpretation pivoted toward a more reliable log- and core-integrated approach.

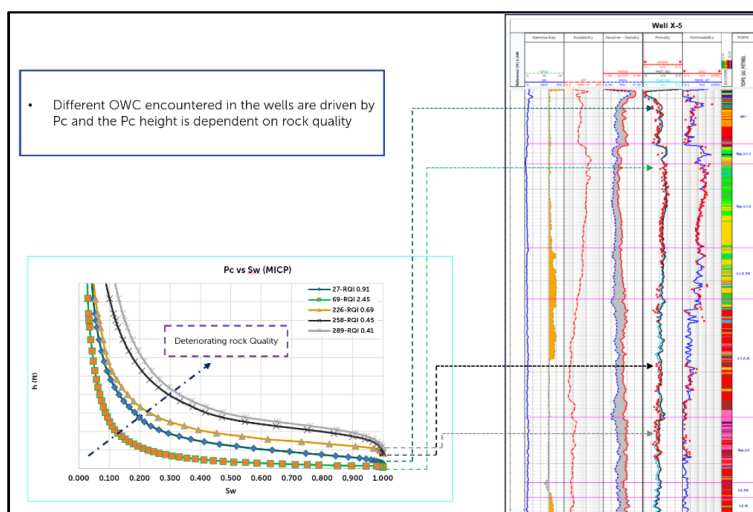


Figure 3—Pc vs Sw for various RQI

Capillary pressure (P_c) data, obtained through laboratory mercury injection capillary pressure (MICP) tests and centrifuge measurements, played a critical role in identifying the underlying control on contact elevation. These data revealed a systematic trend in entry pressures and pore throat sizes, with the Central Area exhibiting significantly higher threshold pressures. This observation suggests that the fluid contact may not be governed by structural equilibrium alone but rather by facies-controlled capillary forces (Kirkham et al., 2010). In this context, the transition from high-permeability, grain-supported facies in the East to more micritic, argillaceous facies in the Central Area creates a natural capillary barrier, enabling oil to reside at shallower depths despite dynamic connectivity.

Furthermore, permeability-log transforms derived from core-calibrated porosity-permeability relationships provided insights into lateral rock quality variation (Figure 4). Facies with poor permeability were more prevalent in the Central Area, further supporting the hypothesis that capillarity and rock texture influence the distribution of fluids. The transition zone, characterized by mixed lithologies and poor sorting, serves as a capillary transition that mimics a tilted contact geometry in log and core data, despite the absence of structural compartmentalization.

Collectively, the petrophysical evidence strongly supports a facies-driven contact variation rather than one induced by structural or hydrodynamic effects. The data suggests that what appears as a tilted contact may instead be a gradational capillary transition, with the oil-water interface being controlled by pore throat distribution, rock fabric, and saturation behavior rather than simple depth or pressure equilibrium.

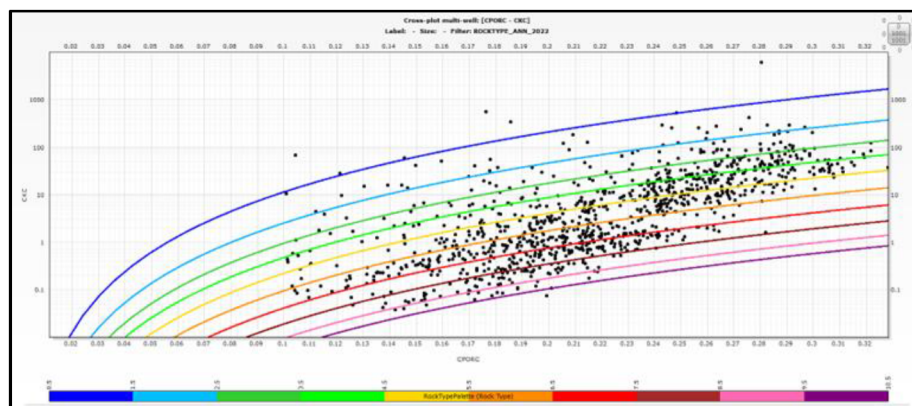


Figure 4—Poroperm Transform

Geological and Geophysical Analysis

A comprehensive 3D seismic interpretation was conducted with the primary objective of investigating whether structural compartmentalization could account for the observed spatial variation in the oil-water contact (OWC) across the field. The seismic data, which covered the full extent of the reservoir, revealed generally continuous and laterally extensive reservoir architecture. No significant through-going faults, sealing barriers, or compartmentalizing structures were identified that could isolate pressure regimes or influence local fluid levels (Figure 5).

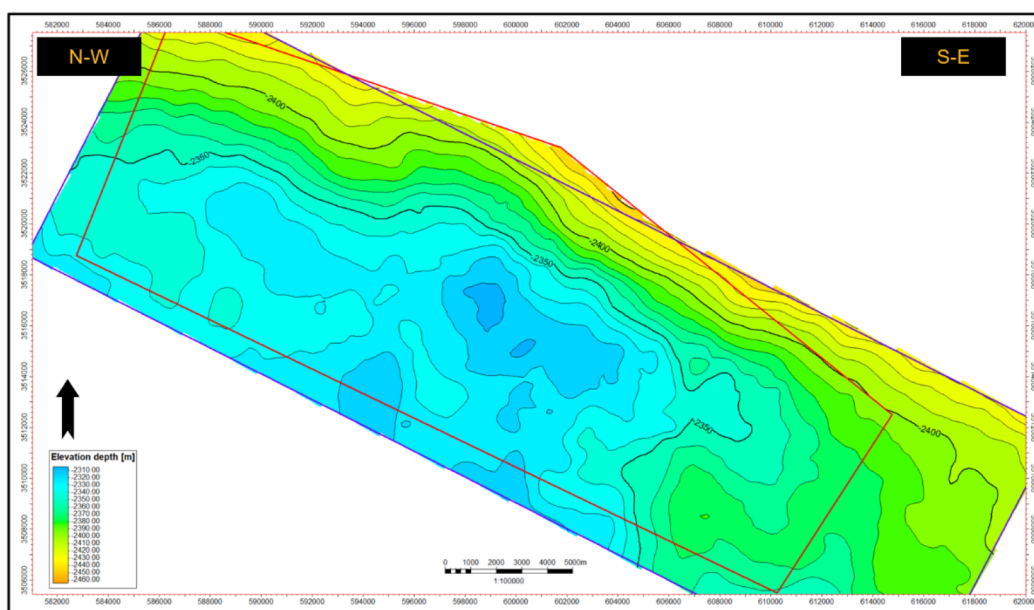


Figure 5—Reservoir X Map Showing No Compartmentalization

Nonetheless, a subtle but persistent structural high was observed in the Central Area. Although this feature does not provide definitive evidence of a controlling structural element, its presence raises the possibility of a long-term tectonic tilt or gentle warping influencing the fluid distribution across the field. This tilt could be a result of post-depositional tectonic activity or differential compaction, which may have subtly altered the fluid contact over geological time.

To further investigate the nature of the contact, sedimentological and petrographic analyses, including thin-section microscopy, were performed (Figure 6). Core samples acquired from multiple wells across the field were utilized to characterize the lithological and petrophysical changes both above and below the interpreted OWC. In the interval above the contact, the reservoir is dominated by grain-supported packstones while the interval below the contact is dominated by argillaceous packstone with calcisphere in parts. These facies are characterized by high interparticle porosity, good grain-to-grain connectivity, and well-developed pore networks, which collectively enhance reservoir quality and promote effective fluid flow. The depositional texture and diagenetic overprints in these zones indicate a high-energy environment, consistent with a shoal or ramp crest setting.

In contrast, the facies below the contact are markedly different, consisting primarily of argillaceous micritic limestones. These rocks exhibit a finer texture with increased clay content, dominated by micrite matrix with isolated intraparticle and microporosity. The overall connectivity in this lower zone is poor, and the porosity is largely occluded or disconnected, leading to a substantial drop in permeability. This abrupt facies transition coincides with a significant increase in capillary entry pressures, effectively acting as a permeability barrier that hinders downward fluid migration. The textural and compositional differences are

also indicative of a shift to a lower-energy depositional environment, potentially associated with more distal or restricted lagoonal settings.

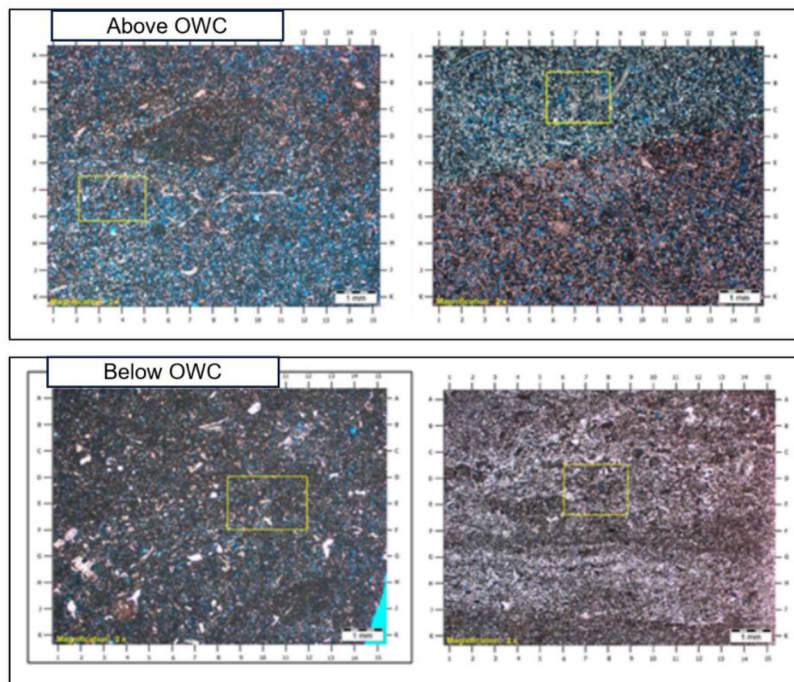


Figure 6—Thin Sections from 2 different Wells Above and Below OWC

Detailed mapping of the contact surface, integrating data from well logs, core descriptions, and seismic interpretation, shows that the OWC geometry aligns closely with the structural top of the reservoir ([Figure 7](#)). This observation is particularly intriguing given the absence of faults or other compartmentalizing features. The OWC exhibits a consistent and gradual dip from the Central Area towards the Eastern flank of the field. This parallelism between the structural top and the fluid contact suggests that the two features may have evolved concurrently, influenced by a combination of depositional processes and long-term structural tilting. The spatial coherence of the contact surface, in the absence of dynamic hydraulic gradients, points to a static equilibrium condition controlled by geological and petrophysical properties rather than by active hydrodynamic flow or pressure differentials.

Additionally, the relatively uniform oil column thickness observed across different parts of the field reinforces this interpretation. The consistency of fluid levels, despite lateral changes in structure and facies, supports the conclusion that the OWC is governed more by stratigraphic and petrophysical heterogeneities than by faulting or pressure segregation. This implies that the field likely represents a single hydraulic unit, with the observed fluid contact shaped primarily by original depositional topography and subsequent diagenetic modification, further enhanced by gentle structural tilt ([Alhamdi et al., 2023](#)).

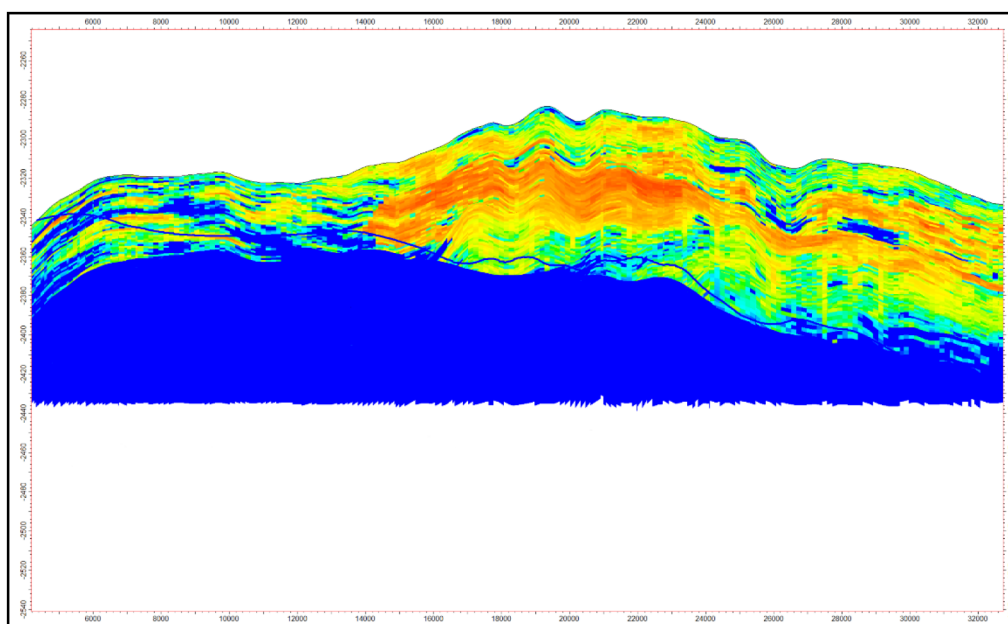


Figure 7—Sw Map with OWC Line (Blue)

Integrated Modelling and Challenges

To evaluate the plausibility of a tilted fluid contact in Field G, a comprehensive 3D static reservoir model was constructed. This model integrated a wide range of subsurface datasets, including interpreted well log data, seismic-derived structural surfaces, facies distributions based on core and log descriptions, and petrophysical properties such as porosity, permeability, and saturation. The initial modelling approach assumed a conventional horizontal oil-water contact, consistent with the traditional view of a structurally uncomplicated and hydraulically continuous reservoir. However, this scenario failed to accurately replicate the observed fluid levels across the field, particularly in the Central and Eastern areas where deviations from a flat contact were consistently encountered in well data.

Subsequently, a revised modelling approach was implemented by introducing a gently inclined, or tilted, contact surface. The modified model produced a significantly improved match with the observed OWC elevations in multiple wells across the Central to Eastern sectors of the reservoir. The spatial alignment between the predicted and observed contact depths under this tilted scenario added credence to the hypothesis that a geological tilt, rather than random measurement errors or compartmentalization, is influencing the contact geometry (Hedayatnejad et al., 2022). This finding marked a critical advancement in understanding fluid distribution within the field and laid the foundation for refining the geological model.

Nevertheless, incorporating the tilted contact into dynamic reservoir simulation introduced a host of technical and conceptual challenges. Historically, Field G has been modeled as a single-tank system exhibiting consistent and strong pressure support, likely maintained by limited natural water influx or pressure connectivity across the field. This modelling assumption inherently implies a uniform contact level, which is incompatible with the presence of two or more fluid contacts. Most commercial reservoir simulation tools are designed to operate under the assumption of either a flat contact or discrete compartmentalization, making it technically challenging to simulate a continuous reservoir exhibiting a tilted fluid interface. In the absence of clear structural barriers, introducing artificial compartments solely to accommodate multiple contacts risks undermining the geological credibility of the model. As a result, the dynamic model struggled to fully honor the contact geometry observed in the static domain, limiting its ability to accurately forecast future reservoir behavior under various production scenarios.

To further constrain the model and reduce uncertainty, additional efforts focused on integrating saturation-height functions derived from capillary pressure laboratory data and well log interpretation. Capillary

pressure curves from representative core samples were calibrated with resistivity logs and formation testers to better estimate fluid saturations across a range of reservoir conditions. Permeability profiles were also examined in detail, revealing a consistent trend of decreasing permeability near the oil-water transition zone in several wells (Figure 8). This reduction in permeability coincided with a lithological shift from clean, well-sorted grainstones and packstones to finer-grained, argillaceous micrites with lower connectivity. The alignment of permeability decline, facies transition and contact position points to a stratigraphically controlled flow barrier that acts in conjunction with the structural tilt to influence fluid segregation.

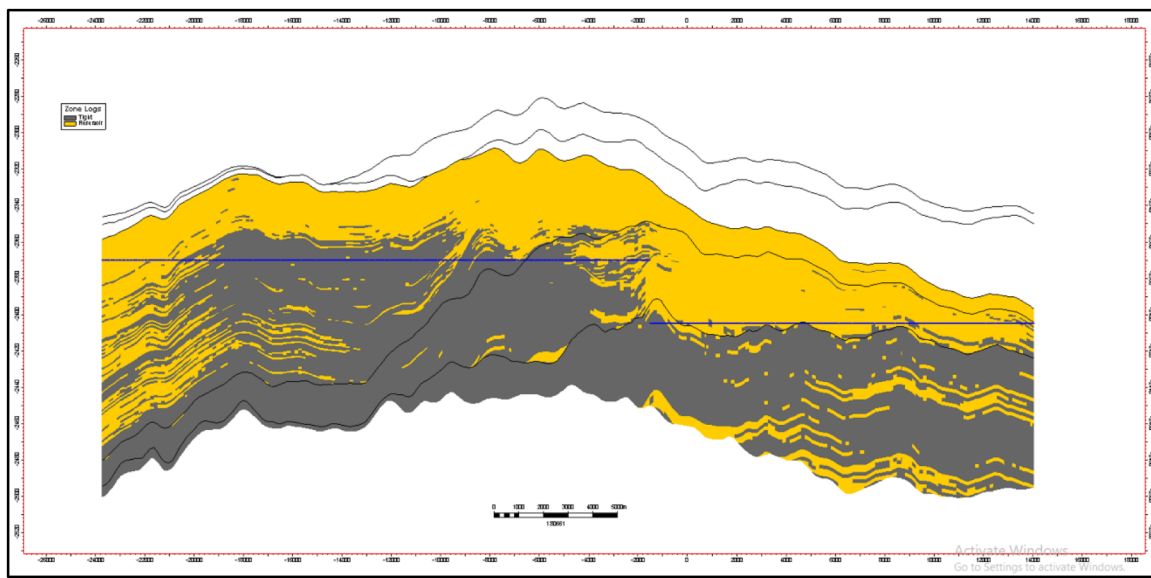


Figure 8—Structure Map with Permeability Distribution with 2mD Cutoff

This facies-controlled transition is not unique to Field G. Comparative studies of other carbonate reservoirs globally especially those with low matrix permeability and strong diagenetic overprint have reported similar phenomena. In these analogues, tilted contacts have been attributed not to active hydrodynamic flow, but to long-term geological processes, such as differential compaction, basin tilting, or preferential cementation of particular facies. In Field G, the limited permeability in the lower reservoir particularly within the water leg along with the absence of a strong external aquifer drive, diminishes the likelihood of a hydrodynamic gradient being responsible for the observed tilt. Moreover, no significant pressure gradient indicative of flow-induced contact movement was observed in formation tester data, further supporting a static contact interpretation.

Ultimately, the close parallelism between the tilted contact and the structural top of the reservoir, as consistently mapped from seismic and well data, reinforces the interpretation that the observed fluid contact geometry is geologically controlled. The most plausible explanation remains a combination of subtle structural tilting and facies-dependent capillary behavior, rather than pressure-driven displacement. This interpretation aligns with established case studies from similar carbonate reservoirs in the Middle East where stratigraphic and diagenetic factors are increasingly recognized as dominant controls on contact geometry and fluid distribution.

While the static model provides a geologically reasonable explanation for the tilted OWC, reconciling this with the dynamic simulation remains a key challenge. Future work may explore hybrid modelling approaches, such as dual contact methods or pseudo-compartmentalization strategies, to better capture the complexity of the reservoir without compromising geological validity. Additional data from extended pressure monitoring, interference testing, and potential fluid sampling in under-drilled areas may also

help refine both the static and dynamic models, ultimately leading to improved prediction of reservoir performance and enhanced recovery planning.

Conclusion

This study presents strong evidence for a facies-controlled, structurally influenced tilted oil-water contact (OWC) in the heterogeneous carbonate reservoir of Field G. The observed variation in contact depth cannot be explained by structural compartmentalization or dynamic aquifer behavior. Instead, integrated interpretation of thin sections, capillary pressure data, resistivity logs, and facies correlations supports a geologically static mechanism, where the fluid contact is primarily governed by changes in rock fabric and petrophysical properties. A clear facies boundary marks the transition from permeable grain-supported lithologies to tight micritic limestones, accompanied by a sharp increase in capillary entry pressure and a shift in resistivity response. This petrophysical contrast acts as a stratigraphic barrier, limiting downward oil migration and giving rise to a tilted contact surface. The mapped contact geometry closely follows the structural top of the reservoir, reinforcing the interpretation of a static, post-depositional tilt preserved over geological timescales.

These findings emphasize the limitations of assuming horizontal contacts in static models, particularly in facies-driven systems. Incorporating tilted contacts enhances volumetric accuracy, improves pay zone delineation, and prevents misinterpretation of flow barriers. Dynamic models must also be refined to reflect such heterogeneities without relying on artificial compartmentalization.

For future development in Field G and similar reservoirs, understanding the interplay between facies, structure, and capillary behavior is critical. Sequence stratigraphic refinement, targeted coring in peripheral areas, and application of machine learning for facies prediction in uncored wells could further improve model reliability. Though not the main objective, this study outlines a scalable workflow that integrates core, log, and seismic data to interpret tilted contacts in complex carbonates.

Ultimately, the insights gained not only refine Field G's reservoir model but also provide a transferable methodology for contact mapping in analogous carbonate systems. Recognizing subtle structural tilts and facies-controlled barriers can lead to more accurate reserve estimates, better sweep prediction, and improved development outcomes in mature and challenging reservoirs.

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References

1. Kirkham et al, 2010, Effect of Tilting on the Capillary Pressure Regimes in Carbonate Oil Rim Reservoirs, GeoArabia Fifth Middle East Petroleum Geoscience Conference, *SPE paper on tilting effects at fluid contacts*
2. Kumar et al, 2002, Saturation Modeling in a Multilayered Carbonate Reservoir Using Capillary Pressure Based Saturation Height Function: A Case Study of the Svk Reservoir in the Y Field, SPE 75213, SPE/DOE Improved Oil Recovery Symposium
3. Alhamdi et al, 2023, The Impact of Cenozoic Tilting on Fluid Contacts and Hydrocarbon Distribution in Upper Shuaiba Carbonates, ADIPEC Technical Proceedings
4. Hedayatnejad et al, 2022, Improved Characterization and Modeling of Capillary Transition Zone in Carbonates for Saturation Height Calibration, REE 191 197247, Reservoir Evaluation & Engineering Conference